

**WEB-BASED DEFENSE SCHEDULING APPLICATION FOR THESIS AND
CAPSTONE PROJECTS AT CITY COLLEGE OF TAGAYTAY**

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WEB-BASED DEFENSE SCHEDULING APPLICATION FOR THESIS AND CAPSTONE PROJECTS AT CITY COLLEGE OF TAGAYTAY

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Chapter 1

INTRODUCTION

Scheduling is the process of planning, organizing, and allocating tasks, resources, and time to ensure that activities are carried out efficiently and without conflict. Scheduling plays a critical role in ensuring the efficient operation of both academic institutions and workplaces. It prevents confusion, saves time, and ensures that people and resources are used effectively. According to Gonzales and Tumang (2020), implementing a centralized scheduling system can significantly reduce faculty workload, eliminate scheduling conflicts, and improve communication between stakeholders. Similarly, Rungta et al. (2021) emphasized that centralized scheduling provides real-time updates, minimizes human error, and enhances overall organizational efficiency. These findings highlight the importance of adopting technology-driven solutions to replace manual scheduling processes in educational settings.

A Web-Based system can be an effective tool for managing various academic processes because it provides accessibility, organization, and centralized control. One important application of this technology is in handling the scheduling of academic defenses, which often requires coordination among students, research advisers, and panel members. To address this need, the researchers will create a Defense Scheduling System for Thesis and Capstone Projects. This system is intended to provide a structured platform that organizes and manages defense schedules in a clear and systematic way. It will help arrange schedules properly, ensuring that participants are assigned to appropriate dates and venues without conflicts.

At City College of Tagaytay (CCT), the existing defense scheduling system for thesis and capstone projects is carried out manually. Faculty members and coordinators rely on spreadsheets, printed forms, and verbal arrangements to set and confirm schedules. Since there is no centralized platform, all details must be managed and communicated personally, requiring additional time and effort from faculty members. This manual method serves as the current approach to organizing defense schedules in the institution.

Findings from interviews with unit research coordinators at City College of Tagaytay (CCT) revealed several issues in the current defense scheduling process for thesis and capstone projects. These include overlapping sessions, limited research adviser availability, and confusion in confirming panel assignments, which are especially evident during peak periods when many students request schedules at the same time. The absence of a centralized system makes it difficult to communicate updates and changes, often resulting in last-minute adjustments and miscommunication among students, advisers, and panelists. Faculty members also reported spending additional hours on coordination, reducing the time available for academic mentoring

and research guidance, while students experience uncertainty and stress due to the lack of timely updates.

To address the recurring issues identified through interviews with the research head and unit research coordinators, the development of a “Web-Based Defense Scheduling Application for Thesis and Capstone Projects at City College of Tagaytay” is proposed. The system is designed to serve as a centralized system where defense sessions can be organized, monitored, and updated in a clear and systematic manner. To ensure smooth operation and accountability, the system incorporates distinct user roles. The System Administrator will configure the system, create accounts for Unit Research Coordinators, assign role-based permissions, and manage essential settings such as academic terms, available rooms, and time slots. Responsibilities also include maintaining data security, managing backups, and generating reports for administrative reference. The Unit Research Coordinators are responsible for managing and finalizing defense schedules by carefully reviewing the proposed schedule submitted by Research Advisers. The Unit Research Coordinator create and manage accounts for Research Advisers and Panelists, assign panel members to specific research groups, and ensure that each defense has the required evaluators. Unit Research Coordinator also confirm the availability of venues and time slots, making sure that schedules do not overlap or conflict with other academic activities. In addition, Unit Research Coordinator validate the accuracy of submitted details and coordinate with both Research Advisers and Panelists to guarantee that all requirements are met before approving the final defense schedule. The Research Advisers are responsible for registering their students’ project groups by inputting complete details such as the research title, list of group members, assigned Research Adviser, and designated Technical Critic/Statistician. Including the Research Adviser and Technical Critic/Statistician in the records is essential to prevent scheduling conflicts, ensuring

they are not assigned as panelists to the groups they directly handle. Research Advisers propose preferred defense schedules based on their group's readiness, while following the time slots set by the Unit Research Coordinators. Then Research Adviser track the status of these requests until they are reviewed and finalized by the Unit Research Coordinators. Through these functions, Research Advisers help maintain accurate records, improve coordination with panelists, and support a fair and well-organized defense scheduling process within the institution. The Panelists are responsible for submitting their availability to the Unit Research Coordinators to help in the preparation of the defense schedule. Once the availability is considered and schedules are finalized, the Panelist wait for the official assignment of defense sessions. Panelists are provided with complete details of the scheduled defense, including the research title, list of group members, and the designated time and defense room, ensuring they are fully informed and prepared to participate in the evaluation process.

Objectives of the Study

The main goal of this study is to develop a system entitled, "Development of Web-Based Defense Scheduling Application for Thesis and Capstone Projects at City College of Tagaytay", that provides a clear and systematic way of managing defense schedules.

Specifically, this study aims to:

1. identifying the system requirements through gathering relevant data from the authorized users to understand the challenges and needs of the current scheduling system.
2. analyze the requirements gathered through the use of different tools and instruments, such as a theoretical framework and data flow diagram.

3. design and develop the system entitled, “Development of Web-Based Defense Scheduling Application for Thesis and Capstone Projects at City College of Tagaytay” that is capable of:

- a. allowing the System Administrator to configure the platform, manage Unit Research Coordinator accounts, assign role-based access, set academic terms, venues, and time slots, while maintaining security, backups, and system reliability.
- b. allowing the Unit Research Coordinators to create accounts for Research Advisers and Panelists, assign panelists, validate schedules, and finalize defense dates while avoiding venue or time conflicts.
- c. allowing the Research Advisers to register project groups with details such as research title, members, research adviser, and technical critic/statistician to prevent conflicts, propose schedules within available slots, and track approval until confirmation.
- d. allowing the Panelists to submit their availability to unit research coordinators and, once scheduled, access complete defense details including research title, research group members, time, and defense room.
- e. supporting systematic scheduling, adjustments, and finalization by organizing availability data from research advisers, panelists, and defense rooms to ensure accurate and conflict-free timetables.
- f. sending timely notifications and reminders to stakeholders about upcoming defenses, approvals, and schedule changes.

- g. generating reports that summarize scheduled and completed defenses, panel assignments, and outcomes for academic and administrative records.
 - h. keeping a detailed activity log of all actions such as account creation, submissions, approvals, and updates to ensure transparency and accountability.
 - i. ensuring data security and reliability through controlled access, encryption, regular backups, and maintenance for continuous system operation.
4. conduct test and validation procedures to ensure the system functions correctly, detects scheduling conflicts, maintains data accuracy, and operates reliably under various scenarios.
5. evaluate the system's performance, usability, and effectiveness using ISO 25010 as the evaluation instrument, based on feedback from authorized users, with emphasis on how well it improves coordination, reduces manual workload, and resolves common scheduling issues.
6. implementation plan of the system, including the setup of user roles, access permissions, and procedures for adding it to the City College of Tagaytay's current workflow for thesis and capstone defenses.

Scope and Limitations of the Study

This study focuses on the design of the system entitled "Development of Web-Based Defense Scheduling Application for Thesis and Capstone Projects at City College of Tagaytay". The system aims to address the issues of the existing manual process, including scheduling

conflicts, miscommunication, and inefficient coordination. By centralizing the scheduling workflow, the system helps ensure that all tasks are completed in a timely and organized manner.

The system is intended for four users, namely the System Administrator (RDE Director), Unit Research Coordinator, Research Adviser, and Panelist. These roles were selected based on institutional requirements and to avoid overcrowding the system's structure. Students are not given direct access to the application. Instead, their information is managed by Research Advisers and Unit Research Coordinators to maintain a simplified access structure and reduce user workload.

The **System Administrator** begins by configuring the platform to ensure it is fully functional for institutional use. This includes creating accounts for Unit Research Coordinators, assigning permissions based on roles, and setting academic terms, available defense rooms, and time slots. The System Administrator also secures the system through controlled access and regular backups while maintaining performance and generating reports for monitoring usage, scheduled defenses, and overall system activities. Once the system is set up, the **Unit Research Coordinator** takes charge of managing the defense process. Creating and manage accounts for Research Advisers and Panelists, assign panel members to specific research groups, and ensure that each defense has the required evaluators. The Unit Research Coordinator also confirm the availability of defense rooms and time slots, carefully preventing overlaps or conflicts with other academic activities. In addition, the Unit Research Coordinator validates the accuracy of submitted research group details and works closely with both Research Advisers and Panelists to make sure all requirements are complete before approving and finalizing defense schedules. The **Research Adviser** uses the system to register their students' project groups by providing details such as the research title, list of members, the assigned Research Adviser, and Technical Critic/Statistician, which prevents conflicts in panel assignments. Research Advisers then propose preferred defense

schedules based on their group's readiness, following the available time slots set by the Unit Research Coordinator. The Research Adviser track the status of the submissions until a final schedule is approved, and once confirmed, they review the assigned panelists, defense room, and time before informing the students. The **Panelists** log in to the system to declare their availability, which becomes part of the scheduling process managed by the Unit Research Coordinator. Once schedules are finalized, Panelists gain access to full defense details including the research title, list of members, time, and defense room so the Panelist can prepare for their assigned evaluations. Together, these coordinated functions ensure that "Development of Web-Based Defense Scheduling Application for Thesis and Capstone Projects at City College of Tagaytay" provides an organized, fair, and efficient scheduling process for thesis and capstone project defenses.

The system is composed of several integrated modules that work together to streamline the defense scheduling process. The **User Management Module** manages account creation, role assignments, and access levels, ensuring that System Administrators, Unit Research Coordinators, Research Advisers, and Panelists have permissions suited to their responsibilities. The **Group Registration Module** allows Research Advisers to register their students' thesis or capstone groups and encode essential details such as research titles, members, and assigned research adviser and technical critics/statistician. The **Group Scheduling Sub-Module** enables Research Advisers to propose preferred defense schedules, which are then reviewed, validated, and finalized by the Unit Research Coordinators. The **Scheduling Module** oversees the setting, adjustment, and confirmation of defense dates, preventing conflicts across defense room, panelists, and academic activities. The **Availability Management Module** is accessible to specific users based on their roles: the System Administrator inputs the availability of defense rooms, the Unit Research Coordinator manages and updates the time slots for these defense rooms, and the Panelists provide

their schedule availability. This coordinated input allows the system to efficiently match panelists, time slots, and venues for each defense. The **Notification Module** provides timely alerts and reminders to System Administrator, Research Advisers, Panelists, and Unit Research Coordinators regarding approved schedules, upcoming defenses, or necessary changes. The **Reporting Module** generates summaries of scheduled and completed defenses, outcomes, and attendance records, supporting evaluation and documentation. To maintain transparency and accountability, the **Activity Log Module** records all actions in the system, including account creation, scheduling requests, approvals, and updates. Lastly, the **System Settings Module** allows System Administrator to configure academic terms, manage courses, update institutional preferences, and ensure that the platform remains aligned with institutional requirements.

However, the system has certain limitations. It is focused solely on scheduling functionality and does not include features like document submission, grading, scoring, or plagiarism detection. As such, all academic evaluation processes must be conducted outside the system. It also does not sync with external calendars or academic timelines, meaning all availability updates must be entered manually by the users. Failure to keep these updated may lead to schedule conflicts. The system requires an internet connection to function, as it does not have an offline mode. Users may experience delays or be unable to access schedules during network outages or unstable connectivity. Although the system is mobile-friendly and accessible on smartphones and tablets, its performance and layout are optimized for modern web browsers such as Google Chrome, Microsoft Edge, Safari. Users accessing the system on outdated browsers, older devices, or with limited processing power may encounter slower loading times, layout inconsistencies, or certain features not functioning as intended.

In summary, this study develops a practical and focused web-based application aimed at resolving common scheduling issues experienced in the current manual defense setup at City College of Tagaytay. While it enhances the coordination process, the system does not aim to cover all aspects of academic project management. Its design intentionally limits user access to reduce complexity and focuses on creating a smooth, accessible, and reliable defense scheduling experience for authorized users only.

Significance of the Study

This study is significant as it introduces a scheduling system designed to make the organization of research defenses more efficient, accurate, and well-coordinated. It aims to reduce errors, save time, and keep important information in one place, ensuring a smoother process for everyone involved. The system is intended to meet the needs of its users and is expected to benefit various individuals within the City College of Tagaytay. Specifically, the results of this study would be valuable to the following sectors:

RDE Director serves as System Administrator. Having centralized control over user accounts, system settings, and schedules. The system's real-time monitoring, activity logs, and centralized reports help them maintain data accuracy and security easily. These tools reduce manual tasks, lower the chance of errors, and allow administrators to manage the system more efficiently and effectively.

Unit Research Coordinators. Use an organized and accurate scheduling process that keeps availability information up to date. Built-in conflict detection prevents double bookings, saving time and reducing scheduling disputes. These features help unit research coordinators assign defense slots fairly and plan more efficiently, leading to smoother overall management.

Research Advisers. Stay informed of defense timelines without the need for constant manual tracking. Receive timely updates on important deadlines and changes, reducing communication gaps and improving time management. That allows Research Advisers to focus more on providing academic support to students.

Panelists. Can benefit from the system by submitting their availability directly, ensuring their schedules are considered in defense planning. Panelist will receive complete details of assigned defenses such as titles, members, rooms, and time slots, keeping them prepared for evaluations. With timely notifications and reminders, panelists stay updated on changes and upcoming sessions, reducing conflicts and miscommunication while allowing them to focus on their role as evaluators.

The Researchers. Gain valuable experience in designing and developing a real-world application that solves a practical problem. Improving skills in system analysis, programming, and project management. Working on this system also allows them to apply technical knowledge, enhance their problem-solving abilities, and prepare for future career opportunities in technology and software development.

Future Researchers. Can build on the existing system to improve and expand its features. They have the opportunity to apply new technologies, explore innovative solutions, and customize the system to better meet user needs. This experience helps them develop advanced technical skills and encourages creativity, preparing them for more complex projects in their careers.

Theoretical Framework

The Theoretical Framework of this study shows what a certain user does in the system.

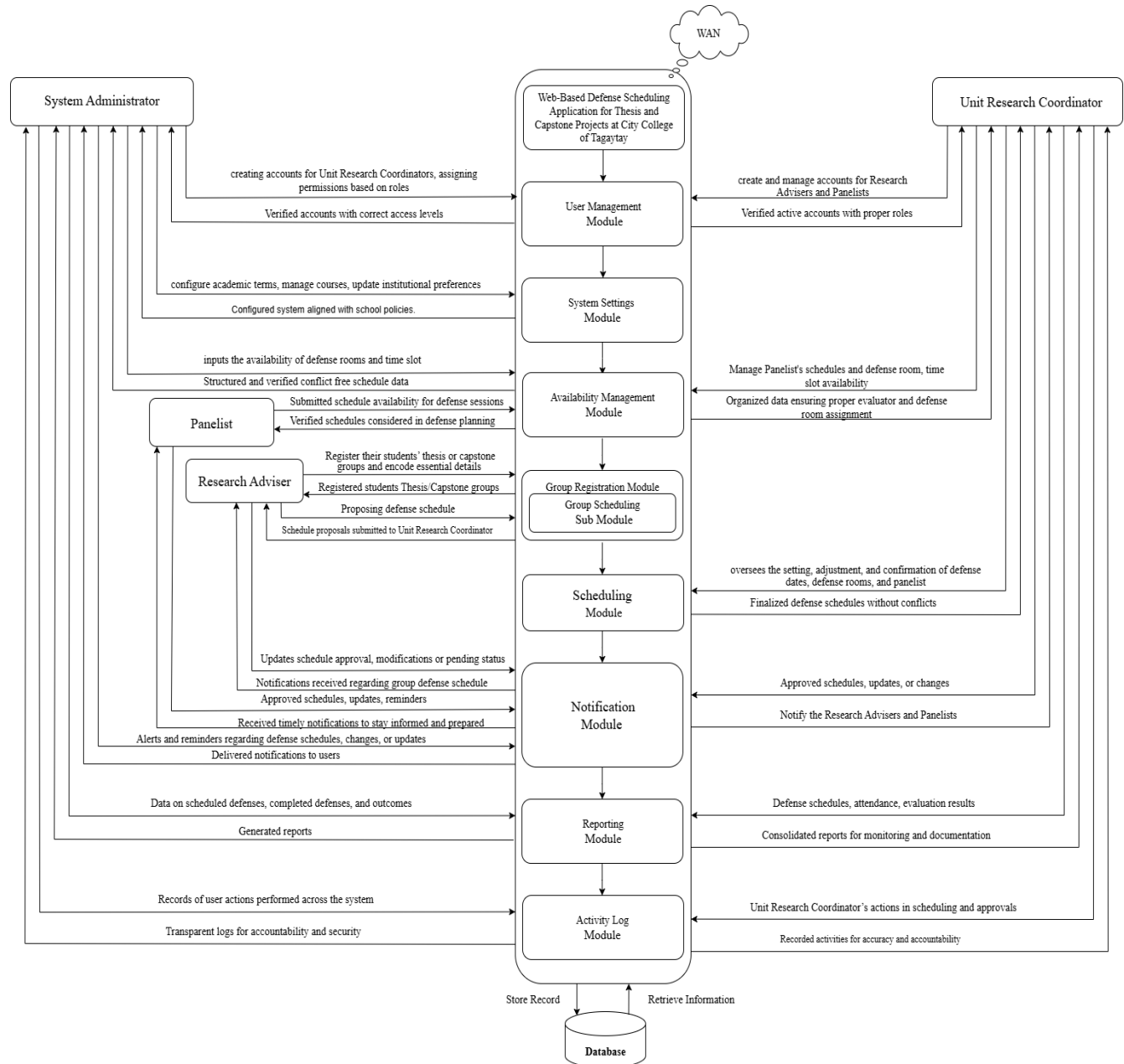


Figure 1. Theoretical Framework of the Development of a Web-Based Defense Scheduling Application for Thesis and Capstone Projects at City College of Tagaytay

The theoretical framework, as illustrated in Figure 1, presents the various modules and single sub-module involved in the Development of a Web-Based Defense Scheduling Application for Thesis and Capstone Projects at City College of Tagaytay. Each module and sub-module interacts with specific stakeholders to ensure the smooth planning, coordination, and monitoring of thesis defense sessions.

User Management Module

This module handles the creation and management of user accounts within the system. It assigns specific access levels to different users based on their roles, such as system administrators, unit research coordinators, and research advisers. By doing so, it ensures that each user can only access the features and data relevant to their responsibilities, maintaining security and proper control over the system.

System Settings Module

This module enables administrators to configure and manage system-wide settings. It covers academic terms, course information, and other institutional requirements, ensuring that the scheduling process aligns with school policies and guidelines. By adjusting these configurations, the module ensures flexibility and adaptability to different academic periods.

Group Registration Module

Through this module, research advisers can register their students' thesis or capstone groups in the system. It facilitates the input of project details such as titles, group members, research advisers, assigned technical critic/statistician, enabling the system to keep track of each group's defense status and project information for easier monitoring and management.

Group Scheduling Sub-Module

As part of the Group Registration Module, this sub-module allows registered groups to propose or request their preferred defense schedules. These requests are then reviewed and finalized by the unit research coordinator to ensure fairness and proper time allocation, preventing overlaps or conflicts with other scheduled defenses.

Availability Management Module

This module gathers availability data from panelists and room resources. Using this information, it assigns appropriate rooms and time slots for each defense, ensuring that conflicts are avoided and that everyone involved, including students, panelists, and system administrators, stays informed about the scheduled defenses.

Scheduling Module

The Scheduling Module is responsible for organizing and managing defense dates. It allows authorized users to set, modify, and finalize schedules while preventing overlapping or conflicting defense sessions. This careful coordination helps maintain an orderly timetable, making sure all defenses proceed smoothly without scheduling issues.

Notification Module

The Notification Module helps keep users informed by sending reminders and updates about upcoming defense dates, schedule changes, or conflicts. It ensures that the research advisers, unit research coordinator, system administrators, and panelist stay updated to avoid missed schedules or misunderstandings.

Reporting Module

This module generates organized reports summarizing defense schedules, results, and other relevant information. These reports serve as official documentation for record-keeping, evaluation, and future reference, assisting administrators and coordinators in monitoring the overall defense process.

Activity Log Module

To maintain transparency and accountability, the Activity Log Module records all actions taken within the system. It keeps detailed logs of user activities and system changes, which can be reviewed to audit the scheduling process and resolve any discrepancies or issues that may arise.

Conceptual Model of the Study

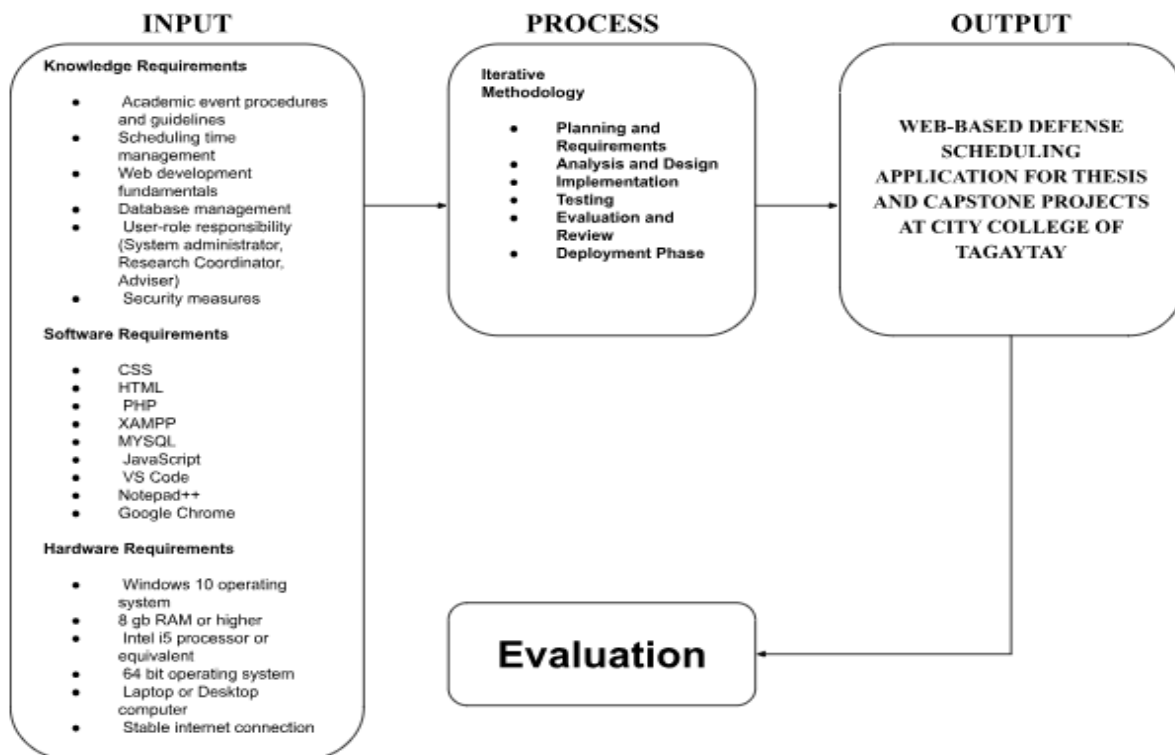


Figure 2. Conceptual Model of the Development of a Web-Based Defense Scheduling Application for Thesis and Capstone Projects at City College of Tagaytay

The conceptual model as illustrated in Figure 2. Shows the different stages of the process involved to achieve the objectives of the study. The input section of the system refers to all the essential resources, knowledge, and tools required for the proper operation and management of the scheduling system. These inputs ensure that users can efficiently interact with the system, enter data correctly, and maintain accurate schedules without errors. The system's performance relies heavily on a combination of user knowledge, appropriate software, and suitable hardware, all of which work together to support its functions. The **Knowledge Requirements** effectively use the system, users need a clear understanding of academic event procedures and scheduling guidelines. This includes knowledge of how defense schedules are set, panelists' roles, and time management for arranging events without conflicts. Users should also have basic web development knowledge, including understanding how websites and databases work, as well as how different user roles such as System Administrators, Unit Research Coordinators, and Research Advisers interact with the system. Knowledge of security measures is important to protect sensitive information and ensure data privacy. In **Software Requirements** several software tools to function properly. HTML, CSS, and JavaScript are used to create the web interface and interactive elements, while PHP handles server-side programming for data processing. MySQL serves as the database management system to store all scheduling information. XAMPP is needed as a local server environment to run PHP and MySQL, while code editors like VS Code or Notepad++ are used for writing and modifying code. Google Chrome or any modern browser is necessary for testing and using the system efficiently. In **Hardware Requirements** a reliable computer is needed to run the system smoothly. The minimum requirement is a Windows 10 operating system with sufficient RAM and storage to handle the software and database operations. A stable internet connection is also important, especially if the system will be accessed online or requires updates. The hardware should be

capable of running multiple applications simultaneously, such as a code editor, browser, and local server environment, without slowing down the system.

The development process follows an iterative methodology, which allows the system to be continuously improved through repeated cycles of refinement. During the **planning and requirements gathering** phase, the needs of the users, such as System Administrators, Unit Research Coordinators, and Research Advisers, are carefully identified, ensuring that the system will address real-world scheduling challenges and avoid conflicts. The system **analysis and design** phase translates these requirements into a clear blueprint, defining how the system will function, how data will flow, and how different modules will interact. During **implementation**, the actual coding and database integration take place, turning the design into a working system. The **testing** phase identifies and corrects errors, ensuring that the system operates reliably and that schedules are accurately managed. **Evaluation** allows feedback from users to be incorporated, making the system more user-friendly and efficient by adjusting features and functionalities based on actual experience. And the **deployment** introduces the fully tested and refined system for real-world use, while the iterative nature ensures that further updates and improvements can be made as new requirements or challenges arise. This iterative approach helps the system remain flexible, functional, and aligned with the users' needs throughout its development and beyond.

The final output is the of Web-Based Defense Scheduling Application for Thesis and Capstone Projects at City College of Tagaytay.

Lastly, the evaluation stage ensures the quality of the system in terms of content, functionality, and efficiency using the ISO 25010 standard. At this stage, the results from **unit testing, integration testing, system testing, and acceptance testing** are analyzed. Unit testing checks individual features like schedule entry, integration testing ensures that modules such as the

scheduling interface and database work together correctly, system testing evaluates the complete system's performance, and acceptance testing confirms that the system meets user requirements. These results help identify areas for improvement, such as fixing errors, optimizing performance, or enhancing the user interface, ensuring the system is reliable, efficient, and ready for deployment.

Operational Definition of Terms

Web-Based Defense Scheduling Application - An online system designed to manage and organize thesis and capstone project defenses, allowing users to schedule, track, and update defense sessions efficiently.

ISO 25010 - An international standard used to evaluate software quality in terms of functionality, reliability, usability, efficiency, maintainability, and portability.

System Administrator (RDE) - The Research and Development Extension Director are responsible for managing and configuring the system.

Unit Research Coordinator - The faculty member for a specific department or school who oversees defense scheduling, assigns panelists, and resolves conflicts. **Research Adviser** - The faculty member assigned to a specific student or group, responsible for guiding their thesis or capstone project and submitting proposed defense schedules to the Unit Research Coordinator for approval.

User Roles - The defined access levels in the system that determine which features and data each type of user (System Administrator, Unit Research Coordinator, Research Adviser) can view or modify.

Panelist – A faculty evaluator who submits availability for scheduling and is assigned to student defenses. Panelists receive complete details of the defense, including project title, group members, time, and venue, to ensure they are prepared to assess and provide feedback.

Technical Critic – A subject-matter specialist, often in methodology, statistics, or technical content, who reviews a student's research for accuracy, validity, and rigor. The Technical Critic ensures the study follows proper standards and provides expert recommendations for improvement.

Statistician – A research evaluator who focuses on the accuracy and appropriateness of the statistical tools, data analysis, and interpretation used in a study. The Statistician ensures that results are valid, reliable, and aligned with proper research methodologies.

CHAPTER II

REVIEW OF RELATED LITERATURE AND STUDIES

This chapter introduces the related literature and studies collected by the researchers from intensive research. It gives valuable information that aids in developing the Web-Based Defense Scheduling Application for Thesis and Capstone Projects at City College of Tagaytay. The chapter consists of an overview of existing ideas, relevant theories, and concepts, and significant technologies connected to the project. It also identifies key terms to assist readers in gaining a deeper understanding of the study. In general, this chapter demonstrates the necessity of blending proven methods and contemporary web-based technologies to overcome the issues of arranging academic defenses.

Related Literatures

Research Defense Management System are essential tools for improving the organization and monitoring of academic defenses such as theses, capstones, and feasibility studies. In their article in the Philippine Journal of Educational Technology, the authors investigated a custom-built platform used at a state university that enabled students, panelists, and coordinators to manage research schedules, submit requirements, and view defense outcomes in real time. The findings indicated a noteworthy decrease in scheduling conflicts and administrative lags. Further, the system enhanced documentation and transparency, enabling academic units to monitor student progress and panelist feedback effectively. Dela Cruz and Ramos noted that such a platform is particularly precious during hybrid or online learning setups, where manual administration of research defenses is liable to miscommunication and error (Dela Cruz and Ramos 2023).

Cloud Computing has significantly enhanced the scalability, accessibility, and performance of web-based scheduling systems in academic institutions. Published in the Journal of Cloud Applications in Education, their study explored how integrating cloud platforms such as Amazon Web Services (AWS), Google Cloud, or Microsoft Azure allows scheduling systems to handle larger volumes of data, enable multi-user access in real time, and support continuous system availability regardless of geographic location. The authors noted that cloud-based deployment eliminates the limitations of local hosting, such as hardware constraints and manual backups, by offering on-demand computing resources and automatic scaling. In their case study involving a university defense scheduling system, cloud integration resulted in improved system reliability, faster response times, and secure data synchronization across departments. Villareal and Santos concluded that cloud computing not only supports the flexible deployment of academic scheduling platforms but also future-proofs them through high availability, disaster recovery, and cost-effective maintenance (Villareal and Santos 2023).

Web Technologies modern web technologies such as HTML, CSS, JavaScript, including AJAX and jQuery, PHP with Laravel framework, and MySQL are essential tools in the development of dynamic and interactive scheduling platforms, particularly in academic institutions. In their study published in the International Journal of Education and Management Engineering, they introduced a Smart Class Routine Management System designed to automatically generate conflict-free timetables for teachers, classrooms, and laboratories. The system utilizes a modular and scalable architecture, making it adaptable across different departments and academic programs. Through real-time data handling and role-based access control, the platform ensures that administrators, faculty members, and staff interact with the system according to their specific roles. The authors concluded that using these technologies

significantly improves user experience, reduces scheduling errors, and minimizes the administrative burden associated with manual scheduling methods, making it a practical solution for institutions aiming to streamline their academic operations (Kabir, and Ahmed 2022).

Sublime Text is a popular code editor widely used by developers for building web-based systems, including academic scheduling platforms. In their study published in the *Philippine Journal of Information Technology and Systems Development*, they highlighted Sublime Text's efficiency, speed, and versatility in supporting web technologies such as HTML, CSS, JavaScript, PHP, and MySQL. The authors emphasized that features like syntax highlighting, multi-line editing, real-time preview through extensions, and project folder management help streamline the development process and reduce coding errors. In their case project, Sublime Text was utilized to develop and test a class scheduling system, allowing developers to rapidly iterate through interface designs and backend scripts in a unified workspace. Navarro and Jimenez concluded that while Sublime Text is not a compiler or server, its lightweight interface and strong plugin ecosystem make it an ideal tool for front-end and back-end development during the early phases of web-based scheduling system creation (Navarro and Jimenez 2022).

CSS (Cascading Style Sheets) play a vital role in enhancing the visual presentation and responsiveness of web-based applications, including academic scheduling systems. In their study published in the *International Journal of Web-Based Learning and Teaching Technologies*, they emphasized that modern CSS, particularly CSS3 enables developers to implement responsive design techniques that adapt to various screen sizes and devices without compromising user experience. Features such as media queries, flexbox, and grid layout have become essential in creating intuitive and mobile-friendly interfaces. The authors noted that in the context of academic scheduling systems, CSS is instrumental in organizing complex visual data such as time blocks,

calendar views, and user-specific dashboards into clear, readable formats. This significantly improves usability for students, faculty, and administrators. Nakamura and Reyes concluded that the use of advanced CSS techniques contributes not only to aesthetic appeal but also to improved accessibility, navigation, and overall system functionality in education-focused web platforms (Nakamura and Reyes (2022)).

PHP (PHP Hypertext Preprocessor) remains one of the most widely used server-side scripting languages for developing dynamic and data-driven web-based systems, including academic scheduling platforms. In their study published in the Southeast Asian Journal of Information and Communication Technology, they explored the implementation of a university scheduling system developed using PHP in combination with MySQL. Their research highlighted PHP's ability to efficiently handle form submissions, session management, and real-time database communication, making it ideal for applications that require user authentication, scheduling logic, and secure data transactions. Additionally, the use of PHP frameworks such as Laravel and CodeIgniter was found to improve development speed, enhance security through built-in protections, and enforce clean coding practices. Mendoza and Tan concluded that PHP provides a stable and flexible backend environment for academic institutions seeking to automate manual scheduling processes and implement role-based access to ensure appropriate control over system functionality (Mendoza and Tan 2022).

HTML (HyperText Markup Language) remains a foundational technology in the development of web-based systems, serving as the structural framework upon which dynamic and responsive user interfaces are built. In their article published in the Journal of Web Engineering and Applications, they emphasized that HTML5, the latest standard, enhances functionality by supporting multimedia elements, form validation, and semantic tags—features that are especially

useful in building user-friendly scheduling systems. These enhancements allow developers to design interfaces that are accessible, lightweight, and compatible across different devices and browsers. In academic scheduling platforms, HTML works in conjunction with CSS and JavaScript to deliver interfaces where users can interact with calendars, forms, and real-time notifications. Liu and Fernandez concluded that HTML's evolution, particularly through HTML5, has significantly contributed to the efficiency and accessibility of educational management systems by enabling structured, interactive, and mobile-responsive design (Liu and Fernandez 2021).

XAMPP is a widely used local server environment that simplifies the development and testing of web-based applications such as academic scheduling systems. In their study published in the International Journal of Software Development Research, they explained that XAMPP is an acronym for Cross-Platform, Apache, MySQL, PHP, and Perl offers an integrated package that allows developers to run and test dynamic websites locally before deployment. The authors emphasized that XAMPP is especially valuable in academic environments where teams build scheduling platforms using PHP and MySQL. It eliminates the need for complex server configurations and allows for rapid prototyping and testing. In their project, which involved developing a class scheduling tool, XAMPP was used to simulate real-time data processing, form handling, and database interactions in a secure and controlled offline environment. Delgado and Cruz concluded that XAMPP is essential for developers during the early stages of system development, helping ensure that web-based scheduling applications are functional, efficient, and error-free before going live on production servers (Delgado and Cruz 2021).

Visual Studio Code (VS Code) is a light yet powerful source-code editor created by Microsoft, commonly used for developing web and software applications. VS Code has support

for various programming languages like HTML, CSS, JavaScript, PHP, and Python and is thus perfectly suited for the development of contemporary web-based systems. It consists of integrated features such as syntax highlighting, debugging, version control using Git, and an extension marketplace where users can personalize the editor for their development requirements. When developing a web-based defense scheduling platform, VS Code offers a clean and productive environment for coding and testing. With its live server extensions, developers are not only able to preview updates in the browser instantly, but that also makes the process of development faster. Due to its versatility, extensive language support, and large community, VS Code has become a highly sought-after tool among both students and professional developers (Potter and Wilson 2021).

Web-Based Scheduling System provides a centralized platform where users can access and manage their schedules at any time and from any location. These systems are designed to enhance accessibility, reduce dependency on physical records, and eliminate common scheduling problems such as human error, overlapping appointments, and delayed updates. By automating the scheduling process, web-based systems promote real-time updates and efficient coordination among multiple users, making them particularly useful in academic and institutional settings, where managing multiple schedules simultaneously is necessary (Sharma 2020).

JavaScript is a scripting language that is utilized to make websites dynamic and interactive. JavaScript enables developers to include functionalities like pop-up windows, form validations, auto-refresh of pages, and adaptive menus. JavaScript plays a critical role in web-based scheduling systems to validate whether users input correct data in forms like dates and times before forwarding it to the server. It is also employed to refresh the schedule in real time without requiring the reloading of the entire page. This results in the system being faster and easier to use.

JavaScript complements HTML and CSS to enhance how users interact with the app, making it a vital element in constructing contemporary web platforms such as a defense scheduling system (Flanagan 2020).

Related Studies

Local Studies

Web-Based Class Scheduling System of College of Information Technology of University of Cagayan Valley

According to Bulan, Cabalza, et. al (2024) of College of Information Technology of the University of Cagayan Valley created a capstone project "Web-Based Class Scheduling System of College of Information Technology of University of Cagayan Valley." The project focused on building an online platform that would help in automating and enhancing the process of class scheduling in the university. Before this system, class scheduling was done manually, and this resulted in usual problems like conflicting class times, room clashes, and inconvenience in the assignment of faculty members to correct time slots. These issues often caused delays and inefficiencies that ended up impacting both students and faculty members. To resolve these issues, the team created an online system that gathers and processes vital scheduling information like subject offerings, instructor availabilities, class sizes, and student block assignments. The system then generates conflict-free schedules by making use of rules and logic embedded in the system. Schedules may also be reviewed and modified by administrators through an easy-to-use interface. By automating the scheduling process, the system minimizes human error, saves time, and facilitates improved interdepartmental coordination. Although the system was originally designed for class scheduling, its characteristics can readily be adapted to other academic activities, including scheduling for thesis and capstone defenses. The capacity to align available time slots

among students, teachers, and panelists, and the convenience of handling it all through a centralized platform, are features that can facilitate numerous academic procedures. This project shows how digital solutions can enable universities to streamline administrative processes, ensure accuracy, and minimize manual work through the application of easy-to-use and well-designed web-based systems.

Web-Based Scheduling and Monitoring Management System

According to Andag, Dionaldo, et. al (2023), a team of Bachelor of Science in Information Technology students from Granby Colleges of Science and Technology in Naic. The capstone project was created to enhance the way the school schedules classes and monitors study activities. Historically, scheduling and monitoring were manually conducted, which was usually prone to errors, overlapping schedules, and ineffective tracking of classes and faculty availability. These problems resulted in delays and unnecessary work for both the administration and faculty. To achieve this, the group developed an online system that automates the scheduling process and supports real-time monitoring of academic schedules. The system enables users to input information such as subject offerings, teacher assignments, and availability of rooms. The system then runs the data to produce accurate and conflict-free schedules. Apart from scheduling, the system also has monitoring features to monitor class activities, attendance, and usage of resources, giving the administration improved visibility and control. This project identifies how digital systems can enhance school operations through minimizing manual work, avoiding conflicts, and increasing access to data. Even if created for class and academic activity scheduling, the features of the system like real-time tracking, data management, and automatic scheduling can also be beneficial in other learning environments. For instance, such functions may be employed in a web-based thesis or capstone defense scheduling application to match availability between students,

advisers, and panelists while tracking event status and logistics. The project shows the promise of harnessing web technologies to enhance efficiency and organization in academic institutions.

Web-Based Thesis/ Capstone Project Defense Evaluation System of the CCS Biñan

According to Garcia, De Leon, et.al. (2023) of the University of Northern Philippines, an online system known as the Web-Based Thesis/Capstone Project Defense Evaluation System was created for the College of Computer Studies – Biñan to enhance the process of thesis and capstone defense assessment and management. Previously, the panelists utilized printed rubrics and manually computed grades while assessing student defenses. This process was not only time-consuming but also subject to human mistakes, misplaced papers, and late provision of feedback to students. The researchers observed these issues and developed a web-based system that accelerated the process, simplified the process, and ensured accuracy. The system enables panelists to directly enter scores into an online form during the defense. These scores are computed automatically and stored in the database of the system, minimizing manual computation and paper work. It also includes student and panelist record management, scheduling facilities, and generating reports. This implies that all aspects concerning the defense e.g., the setting of the schedule, the allocation of panelists, and result viewing can be conducted within one program. The system also ensures that sensitive information is only accessible to authorized users, thereby protecting the privacy of both evaluators and students. Once tested, the system proved to be effective in enhancing the defense process. The instructors testified that the platform minimized their workload and enabled them to deal with several defenses more effectively. Students equally enjoyed quicker results as well as better feedback. The study confirmed that a digital system can enhance not only the assessment process, but also the organization and conduct of thesis and capstone defenses in general. This study validates that online systems are beneficial in academic

environments, particularly in the management of thesis and capstone projects. With the process automated and sorted out, universities can save time, prevent errors, and make both the faculty and the students' experience better during one of the most significant aspects of their educational process.

THESISIT: Web-Based University Thesis Management Portal With A Defense Scheduling System

According to Chio et al. (2022) of the University of the Cordilleras created a system titled THESISIT, a web portal aimed at facilitating the management of thesis processes, particularly thesis defense schedules. The research aimed to address typical issues of defense scheduling, such as duplicate appointments, out-of-date panelists, and tardy coordination. It features the uploading of thesis papers, monitoring progress, and scheduling defenses. One of its strongest points is the employment of a genetic algorithm that automatically generates defense schedules based on panelists' and students' availability and room assignments. This prevents conflicts and saves time versus manual scheduling. The system was developed with the Waterfall development model and tested according to international software standards (ISO/IEC 25010). The results indicated that THESISIT was 84% functional and 94% usable, meaning it could function properly and users liked to use it. It also accommodates various user roles; students, advisers, panelists, and admins each with their own set of permissions and responsibilities. Students can upload documents, advisers can approve proposals, panelists can see defense details, and admins can manage schedules. Overall, the research demonstrates that THESISIT facilitated thesis management to be quicker, better organized, and less stressful for students and instructors alike. It is an excellent model for schools that seek to enhance how they manage thesis and capstone project scheduling online.

An Implementation and Evaluation of Web-Based Appointment System for the Mindanao State University - Main Campus

According to Chua et al. (2022) from Mindanao State University – Main Campus developed a Web-Based Appointment System to improve how appointments are handled at the Office of the University President. Before the system was created, the office used paper slips to schedule appointments, which often got lost or caused confusion. This led to long waiting times, double bookings, and poor tracking of schedules. To fix these problems, the researchers built an online system that allows users to set, check, and manage their appointments through a website. The goal was to make scheduling faster, easier, and more organized. The system was tested using the System Usability Scale (SUS) and the Technology Acceptance Model (TAM) to check how useful and user-friendly it was. The results showed a 90.2% satisfaction rate, which means most users found it helpful, easy to use, and convenient. Staff and clients no longer needed to go to the office in person just to book or confirm appointments. Instead, everything could be done online, saving time and effort. The system also helped the office staff monitor appointments and reduce missed schedules. This study shows that even a simple web-based system can greatly improve how scheduling is done in schools. Although it was made for office appointments, the same idea can be used for thesis and capstone defense scheduling. Features like online booking, schedule tracking, and role-based access can help schools make their academic scheduling more efficient and organized.

A Multi-Objective Model for Thesis Defense Scheduling

According to Almeida, dos Santos, et. al. (2022) proposed an intelligent method for thesis defense scheduling through a mathematical model. They observed that exam and class schedules are usually well-planned, yet thesis defense schedules are neglected, though they could be as complex. To address this, they developed a system based on mixed-integer linear programming

(MIP), one of the advanced math methods for resolving tricky scheduling issues. Their model was designed to be general and adaptable, so that multiple different universities could utilize it without having to accommodate one particular school's requirements. The system they developed assists in determining the optimal defense schedule by balancing a number of key objectives. These are ensuring that faculty members receive an equal workload, preventing too much shifting of room assignments, maintaining compact schedules (near to each other in time), and attempting to accommodate preferred time slots. In order to accomplish this, they employed a two-step approach. First, the system first verifies how many defenses can practically be scheduled within the constraints of available times, individuals, and rooms. Next, it determines the most optimal methods to schedule them and provides many just and effective schedule alternatives for the administrators to select from. The authors validated their model with 96 test cases, representing various actual scenarios, and the results indicated that the system could always obtain a good conflict-free schedule. Their research demonstrates that applying optimization models can assist universities with thesis defense scheduling in a more equitable and efficient manner. This type of strategy could be extremely beneficial for academic institutions who want to minimize scheduling issues, save time, and optimize the whole coordination of thesis defenses. It also indicates that even the most complex processes can be made simpler through the assistance of the correct technology and planning tools.

Web-Based Scheduling System for Educational Institution with QR Code and Facial Recognition

According to Carawana, Delos Reyes, et. al. (2022) from Southern Leyte State University. The primary objective of this project was to enhance the management of school timetables and teacher attendance through the utilization of advanced technology. In most schools, the conventional method of monitoring attendance relies on paper records or manual forms, leading

to lost documents, errors, and time-consuming procedures. To address these issues, the researchers designed an online system that employs both facial recognition and QR code scanning to improve the process's speed, accuracy, and security. Each teacher is assigned a distinct QR code and enrolled in a facial recognition database in their system. Teachers either scan their QR code or utilize facial verification in order to check in for their schedule or verify attendance. Once authenticated, the system logs their attendance and shows them their schedule in real time. This eliminates fraud, minimizes errors, and makes the process paperless. The system also updates schedules in real time and creates reports on attendance, which allows school administrators to monitor and manage activities more efficiently. This method highlights how the combination of simple technologies can have significant effects on daily school work. Although the system was designed for class tracking and attendance, the same principle can be implemented for thesis and capstone project defense scheduling. Through QR code and facial recognition, universities can confirm the attendance of students and panel members, prevent scheduling conflicts, and generate accurate reports for academic records. In general, this 2022 study offers a smart and effective solution that accommodates contemporary scheduling requirements in education.

DefendEase: Automated Pre-Defense Scheduling System

According to Descalso, Laida, et. al. (2022) of Camarines Sur Polytechnic College considered resolving the prevalent problems encountered in scheduling pre-defense sessions for capstone projects and thesis. Scheduling pre-defense in most educational institutions is manually done, thus resulting in delays, conflicts on availability, and excessive administrative effort. These issues are exacerbated when several students and teachers have to synchronize their time with few resources and communication tools. To meet these challenges, the researchers came up with an online automatic system where students are able to make online requests for defense timetables.

Panel members and advisors can feed in their free time slots, and the system will automatically cross-match student and panelist/adviser availability to produce a conflict-free timetable. The system assists in reducing human error and conflicting timetables while enhancing efficiency and speed in pre-defense bookings. It also entails functionalities like reminders, schedule tracking, and approval status updates that keep everyone updated in real time. The system significantly minimizes the effort and time required to schedule pre-defense sessions, thus making it quicker, easier, and more streamlined for both students and staff. The study established that automating the scheduling process not only improves productivity but also aids transparency and improved time management among academic departments. Such a system is a useful example that can be followed by other institutions that want to automate and enhance their thesis and capstone project processes using technology.

A Web Ecosystem for User-Driven Healthcare Appointment Management

According to Maluyo, Osorio, et. al (2022) from Lyceum of the Philippines University – Manila developed a project entitled "A Web Ecosystem for User-Driven Healthcare Appointment Management." It was developed to help patients in the Philippines in simpler appointment reservation and management with doctors. Before this system, many patients had long waiting hours, discontinuous schedules, and poor coordination when they attempted to book appointments in hospitals and clinics. The researchers developed an online system wherein users are able to register, look for doctors by specialty, and view available time slots for consultation. It also had a messaging component so patients can contact health practitioners for immediate inquiries and clarifications. There were two systems: one for users (patients and doctors) and another for the administrators who manage appointments and records. Patients could simply choose the doctor they needed and choose an appropriate time, while doctors could revise their schedules real time.

The system even reminded the patients for their appointments and reduced the need to visit the office just to schedule or confirm appointments. In the test when the researchers tested the site with a subject sample of users, the result showed high satisfaction levels. Individuals found the system convenient, easy to use, and useful. Some recommended enhancing the design and scheduling features, but generally, the system was regarded as a good healthcare appointment management tool. This research indicates how web-based systems can simplify the task of scheduling, make it quicker, and more organized. While it was intended for use in healthcare, the real-time scheduling functionality of the system, user account, reminders, and communication capabilities can also be extended to other industries, such as academic scheduling. For example, in universities and schools, the same type of platforms can be used to handle thesis and capstone project defense by allowing students to schedule a time slot, receive reminders, and communicate with panelists or advisers. It shows that solutions through computers are actually capable of solving real-life problems by making processes more efficient and accessible.

iVacTrack: A Web- based Scheduling System for Covid-19 Vaccination Sites with Patient Tracking Through Geo mapping

According to Cortez and Dizon (2021) from Lyceum of the Philippines University – Manila created a system named iVacTrack, a web-based scheduling system particularly implemented to aid COVID-19 vaccination. The primary purpose of the system was to assist health workers and administrators in efficiently arranging vaccination schedules and monitoring patients' movements through geo-mapping. The system comprised two critical modules one for administrators and another for staff. The administrator module helped health officials design and allocate special time slots for vaccination, preventing overcrowding and confusion. The staff module, however, introduced tools for patient flow and location monitoring in real time through GPS or map-based technology. This aspect guaranteed that health measures, including social distancing, were upheld

when undertaking the vaccination. iVacTrack's capability to integrate scheduling with real-time location monitoring facilitated easy handling of huge masses of individuals, minimized waiting time, and ensured the general safety and management of vaccination sites. It also presented electronic records that could be accessed by both patients and staff for reference, which enhanced the whole process to be made more transparent and dependable. Though the system was meant for health purposes, its architecture and functionality can also be used in other sectors too particularly in academic environments such as thesis and capstone defense scheduling. For instance, just as an appointment is made for vaccination to prevent congestion, thesis defenses can be arranged such that it does not overlap, ensures availability of panelists and students, and maintains the process streamlined and perfectly organized. The research shows how technology can effectively address genuine real-world coordination problems and how such solutions can be implemented in education and other fields.

Foreign Studies

Exam Scheduling Under Pandemic Conditions: A Mathematical Model and Decision Support System

According to Zehra Kamisli Ozturk et al. (2024). The scheduling of university exams is a complex task that involves various constraints such as administrative limits, pedagogical needs, student volume, and different courses. The emergence of Covid-19 and future pandemics has added new constraints related to infection prevention and contact tracing. To address these challenges, this study proposes a multi-objective mathematical model that considers university resources, reduced classroom occupancy, and minimized student interaction. The model aims to minimize violations of pandemic-related constraints and categorize exams by difficulty. To facilitate scheduling for entire faculties or universities, a Genetic Algorithm based web-based

decision support system is developed. With these tools, the study successfully created an optimal schedule for eight departments simultaneously.

An Implementation of Digital Platform to Enhance the Appointment Scheduling System

According to Lim and Teo (2024) from Uversiti Teknologi MARA (UiTM) in Malaysia underscores the necessity of technology solutions for enhancing appointment scheduling systems. Their research points out that traditional appointment management tends to result in scheduling conflicts, long wait times, and a labor-intensive administrative burden. With the introduction of a technology platform, organizations can mechanize the scheduling process, eliminate human errors, and extend prompter and more efficient service to their customers. Lim and Teo employed a combination of methods to gather information, such as interviews and surveys conducted among staff and clients. The findings indicated that the former manual system was complex and error-prone, which generated dissatisfaction among employees and clients. To help solve these issues, the researchers created a digital platform with a simple interface, grouped service categories, and automated scheduling capabilities. Once the system was put into place, the organization experienced a significant reduction in wait time for customers and more efficient workflow for the employees. Not only did the new platform streamline operations, but it also enhanced customer satisfaction. This research concurs with other studies demonstrating the advantages of digital resources in administrative work. For instance, Abu Bakar and Md Said (2019) designed an online booking system for a gym that simplified bookings and improved them for users. Akhtar (2022) designed an online scheduling management system that improved schedule organization and reduced the work burden for staff. All together, these researches indicate how crucial it is to implement digital platforms for appointment booking. Organizations can optimize the way they function, eliminate errors, and offer improved services using technology. These kinds of systems

are particularly useful in locations where handling numerous appointments manually is difficult and time-consuming.

Development of Scheduling System With Genetic Algorithm in Website-Based Smk Negeri1 Sine

According to Shafa Bani Saputra and Endang Wahyu Pamungkas (2023). Scheduling is information that has limited conditions that must be met. Preparation of the schedule will take quite a long time if it is done using conventional media such as writing on paper or books. Scheduling optimization is needed to provide effectiveness and efficiency so that the implementation of learning activities can run more optimally. The genetic algorithm approach method is used to get the optimum schedule. This algorithm produces the best combination for subject pairs and teaching teachers as a whole by determining the initial population and initializing the chromosomes, determining the fitness value, then carrying out crossover selection, and carrying out mutations to produce the best fitness value which will be used to determine the final value of scheduling. The results of the entire algorithm process are consistent with the original prediction data, and the same teacher is not scheduled to teach more than once at the same time. The results of the subject scheduling process using the genetic algorithm obtain a fairly good optimization in subject scheduling.

Web-Based Honey Harvest Scheduling in The Muaro Botuak Jaya (MBJ) Group

According to Nuraeni and Harry (2023) from Ekasakti University Padang. This research is motivated by the mechanism of the conventional honey harvesting system. The harvest schedule is carried out irregularly, resulting in honey that is less than the quality and quantity. Usually this problem can be resolved with an integrated information system that is effective and efficient in managing harvest scheduling. Systems Development Life Cycle (SDLC) SDLC method with a

waterfall model approach. The design tools used in this information system are UML, which consists of use case diagrams, class diagrams, sequence diagrams, activities, collaboration diagrams, implementation. The application used is My SQL database and the PHP programming language. The design of this system produces an information system in the form of a honey scheduling website for the Muaro Botuak Jaya group which was previously written manually to carry out scheduling and with this website the MBJ group can carry out scheduling more effectively and efficiently. Web-Based Honey Harvesting Scheduling System in the Muaro Botuak Jaya (MBJ) Group was built to overcome scheduling problems in the Muaro Botuak Jaya (MBJ) Group, Design of a Web-Based Honey Harvesting Scheduling System in the Muaro Botuak Jaya (MBJ).

Web-based employee work scheduling information system design at pt trafoindo power indonesia

According to Ade Setiadi, Yusuf Effendy et al. (2022). This study aims to assist PT Trafoindo Power Indonesia in improving the description of the employee work scheduling website. So far, several departments of PT Trafoindo Power Indonesia use an application called Microsoft Excel to process employee work schedules and save a file on a public server where each employee has access to open and perform data engineering. The solution to the indication of the problem is to create a new system design that functions to process data in the form of employee work schedules with a one-way input system. The website is one of the choices that companies need if they want to present information both internal and external. A website would be better if there is a database or what is commonly known as a data warehouse, for the basis of reference for decision-makers tomorrow.

In this case, the design of an employee work scheduling information system using Bootstrap as its platform. To obtain the data needed in conducting research, several methods were

used, including interviews, observation, and literature study. The data obtained were then analyzed and described using the UML (Unified Modeling Language) method. The final result of the research is the design of a Bootstrap-based employee work scheduling information system using Visual Studio Code as the code editor.

Design of Intelligent and Secure Hospital Appointment Scheduling System

According to Adamu Muhammad Noma et al. (2022). As a roadmap to digitizing the healthcare delivery system in Nigeria to support effective decisions in healthcare management through harnessing and timely delivery of reliable data, we propose a design of an intelligent and secured appointment scheduling system. The main aim is to support healthcare administration in hospitals, while satisfying the global best practice in healthcare delivery. The system negotiates appointment schedules with medical teams in such a way that it is completely automatic, and optimally satisfies the patients' demands for timely accessing medical service while efficiently utilizing healthcare personnel. With the system, patients have the option of seeking appointments in the comfort of their homes or through record officers in the hospitals. It can handle both daily basis walk-in patients and those seeking appointments with specialists. The system is also designed to be secured at the application layer through a fine-grained group-based access control on authenticated users. When implemented, the system would exist as a subcomponent of the Health Management System and be interoperable with the other components by communicating through standard data exchange formats such as JSON.

Samplify: A user-friendly smartphone app and web-based means of scheduling and sending mobile notifications for experience-sampling research

According to Yury Shevchenko, Tim Kuhlmann (2021) et al. Undertaking an experience-sampling study via smartphones is complex. Scheduling and sending mobile notifications often

requires the use of proprietary software that imposes limits on participants' operating systems (whether iOS or Android) or the types of questions that can be asked via the application. We have developed an open-source platform, Samply which overcomes these limitations. Researchers can access the entire interface via a browser, manage studies, schedule and send notifications linking to online surveys or experiments created in any Internet-based service or software, and monitor participants' responses all without the coding skills usually needed to program a native mobile application. Participants can download the Samply Research mobile application for free from Google Play or the App Store, join a specific study, receive notifications and web links to surveys or experiments, and track their involvement. The mobile application leverages the power of the React Native JavaScript library, which allows it to be rendered in the native code of Android and iOS mobile operating systems. We describe Samply, provide a step-by-step example of conducting an experience-sampling study, and present the results of two validation studies. Study 1 demonstrates how we improved the website's usability for researchers. Study 2 validates the mobile application's data recording ability by analyzing a survey's participation rate. The application's possible limitations and how mobile device settings might affect its reliability are discussed.

Design of a Web-Based Lecture Scheduling Information System During Pandemic Covid-19

According to Novita and Yuggo (2021) from Ibn Khaldun University. During the Covid-19 pandemic, the lecture process was carried out online, so it impacted other academic activities such as the preparation of lecture schedules. The results of observations at the Faculty of Engineering and Science found that the practice of lecture schedules was carried out manually, such as the schedule coordination process was carried out face-to-face between study programs, faculties, and lecturers to overcome conflicts in the use of rooms and teaching time. Changes in

the teaching schedule need to be re-checked on the use of the room and the lecturer's teaching time because it has not been documented with the information system. Hence, this study aims to build an information system for preparing lecture schedules using the Greedy Best First Search Method based on the willingness of lecturers to teach. The system was developed using the RAD (Rapid Application Development) and testing using BlackBox testing. The results of this study succeeded in building a lecture scheduling information system that was able to generate lecture schedules automatically and quickly without having to coordinate face-to-face to support online lectures during the Covid-19 pandemic.

A Web Deployed Multi-Agent Based Approach for Student-lecturer Appointment Scheduling in Institutions of Higher Learning

According to Anthony Bokolo, Gbepa Peremoboere Maureen et al. (2021). Institutions of higher learning such as universities have been positively influenced by the internet which has facilitated learning and teaching. Likewise, this same internet can also support student lecturer appointments, but currently students and lecturers are faced with issues such as finding free timeslot, difficulty of students to meet supervisor and issues related to managing appointment operations. Therefore, this paper develops a multi-agent architecture and a web-based agent appointment scheduling system to support students and lecturers in managing appointment scheduling in universities. The agent-based appointment scheduling system was implemented as a web system integrated by multi-agents to facilitate students in finding available timeslot, resolving difficulty of students to meet supervisor and addressing difficult of managing scheduled appointment records by lecturers. The usefulness of the agent-based appointment scheduling system was evaluated by collecting data using questionnaire from randomly selected respondents. Furthermore, Statistical Package for the Social Science (SPSS) was employed to analyse the questionnaire data using descriptive and exploratory factor analysis. Findings from the survey

reveal that the developed agent-based appointment scheduling system practically supports students and lecturers in scheduling appointments.

Web-Based Appointment Scheduling System

According to Guzalo Noori (2021) of Central Washington University created a web-based Appointment Scheduling System under an undergraduate honors thesis project. The objective of the system was to assist organizations, including schools, clinics, and offices, in booking their appointments more accurately and efficiently. Prior to such systems, appointment scheduling was often conducted manually, and in many cases, this resulted in issues like double bookings, lost appointments, and excessive waiting periods for clients. Noori's system aimed to solve these issues by providing a digital platform where users could easily check available time slots and book their appointments online without needing to call or visit the office. The system was designed to be user-friendly, with simple navigation and clear instructions, so that users with different levels of computer skills could use it without difficulty. It provided real-time updating, such that when an appointment was reserved, available times would be automatically readjusted to avoid clashes and overlaps. This made it possible for no two members to book the same time simultaneously, minimizing errors and misunderstandings. Besides reservation, members were given automatic notifications or confirmations, which assisted in reminding them of their scheduled appointments and minimized the risk of forgetting or missing their appointments. Noori also concentrated on administrators' needs by incorporating management, viewing, and easily adjusting appointments in the tools. This minimized their workload and enhanced overall appointment management. It helped in saving time as evidenced in the study and also made the entire system efficient and more reliable. Users felt less frustrated since they were able to book appointments at any time and from any location, while administrators were able to deal with more appointments with fewer mistakes.

This thesis emphasizes the role played by computerized appointment scheduling systems such as the one created by Noori in the process of modernizing administrative work within various sectors. Through the ease of booking an appointment and enhancing communication between users and administrators, these systems enhance the convenience of daily operations, minimize confusion, and enhance satisfaction. These concepts and technologies can also be applied to other scheduling requirements, like scheduling for academic defense, medical appointments, and event planning, demonstrating the utility of web-based scheduling tools in numerous contexts.

Chapter III

METHODOLOGY

This chapter describes the materials, methodology, project design, and testing procedures.

Project Design

The system entitled “Development of Web-Based Defense Scheduling Application for Thesis and Capstone Projects at City College of Tagaytay” was developed to streamline the scheduling process for research defenses, supporting system administrators, unit research coordinators, and research advisers in managing and organizing defense dates efficiently. The system provides a structured and systematic approach that minimizes scheduling conflicts and manual errors by ensuring clear processes and accurate records. It was built using CSS for layout and design, PHP and JavaScript as server-side scripting languages, and MySQL/XAMPP as the database engine to deliver a reliable, organized, and user-friendly platform.

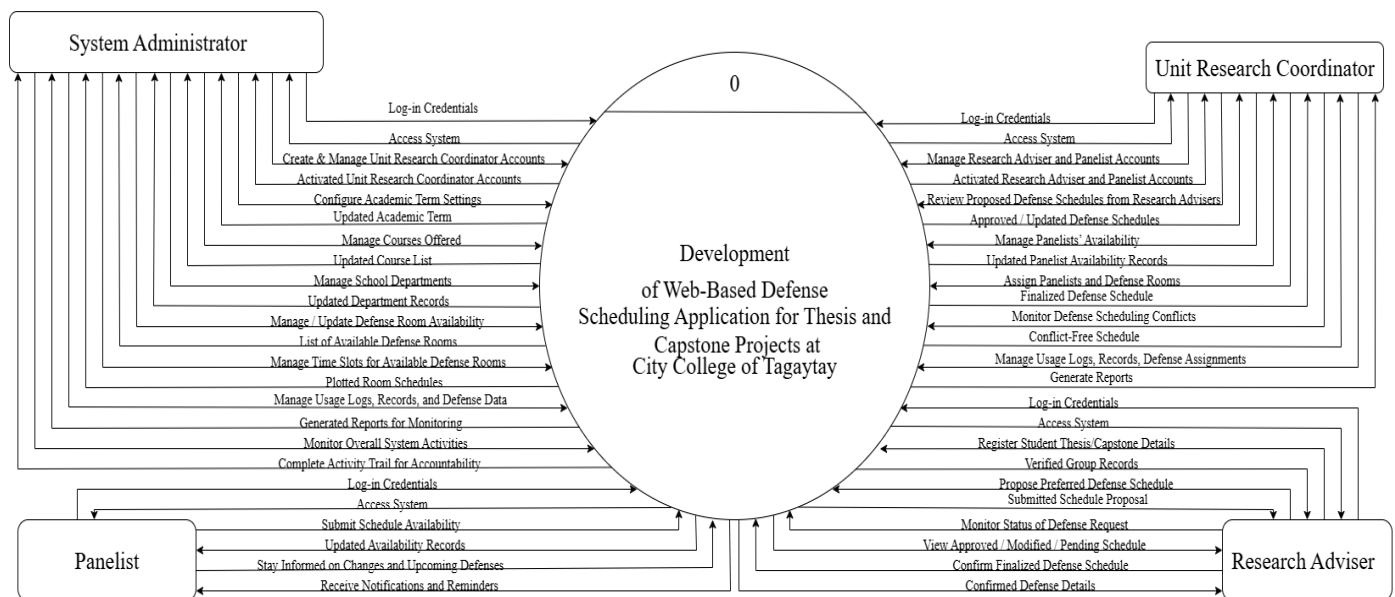


Figure 3. Context Diagram of Web-Based Defense Scheduling Application for Thesis and Capstone Project at City College of Tagaytay

The context diagram in Figure 3 shows the interaction between the external entities and the system. Its main purpose is to help users understand the scope, features, and flow of the system entitled the “Development of Web-Based Defense Scheduling Application for Thesis and Capstone Projects at City College of Tagaytay”. The four (4) entities shown in the diagram are the System Administrator, Unit Research Coordinator, Research Adviser, and Panelist.

The **System Administrator** configures and maintains the defense scheduling system. Their System Administrator responsibilities include creating accounts for Unit Research Coordinators, assigning role-based permissions, and managing system settings such as academic terms, available rooms, and time slots. The System Administrator also ensure data security, maintain system backups, and oversee reports that provide insights for administrative use. Additionally, monitoring overall system activities and generate records to maintain transparency and accountability. The **Unit Research Coordinator** manages and finalizes the scheduling process. They review proposed defense schedules submitted by Research Advisers, create and manage accounts for both Research Advisers and Panelists, and assign panel members to specific research groups. The Unit Research Coordinators are also responsible for confirming defense room and time slot availability, ensuring schedules do not overlap with other academic activities. They validate the accuracy of submitted information and coordinate with both Research Advisers and Panelists to make sure all requirements are met before approving the final defense schedule. Through these functions, they guarantee a conflict-free, organized, and accurate defense schedule. The **Research Adviser** registers student project groups into the system by providing complete details such as the research title, group members, assigned Research Adviser, and designated Technical Critic/Statistician. This prevents scheduling conflicts by ensuring that Research Advisers and Technical Critics/Statistician are not assigned as panelists to their own students. The Research Adviser

propose preferred defense schedules based on group readiness while following the framework provided by the Unit Research Coordinators. The Research Advisers also track the status of defense requests until they are reviewed and finalized. By maintaining accurate group records and coordinating closely with Panelists and Coordinators, they help ensure fairness, transparency, and proper preparation for student defenses. The Panelist contributes by submitting their availability to the Unit Research Coordinator, which becomes essential for preparing conflict-free schedules. Once their availability is considered and assignments are finalized, Panelists receive official details of their scheduled defenses. These include the research title, list of group members, assigned time slot, and designated defense room. With this information, Panelists can prepare ahead of time and focus on their role as evaluators during the defense, ensuring an efficient and well-organized evaluation process.

Together, these four roles establish a coordinated workflow, ensuring that the scheduling of defenses is secure, transparent, and efficiently managed from system administration to student project preparation.

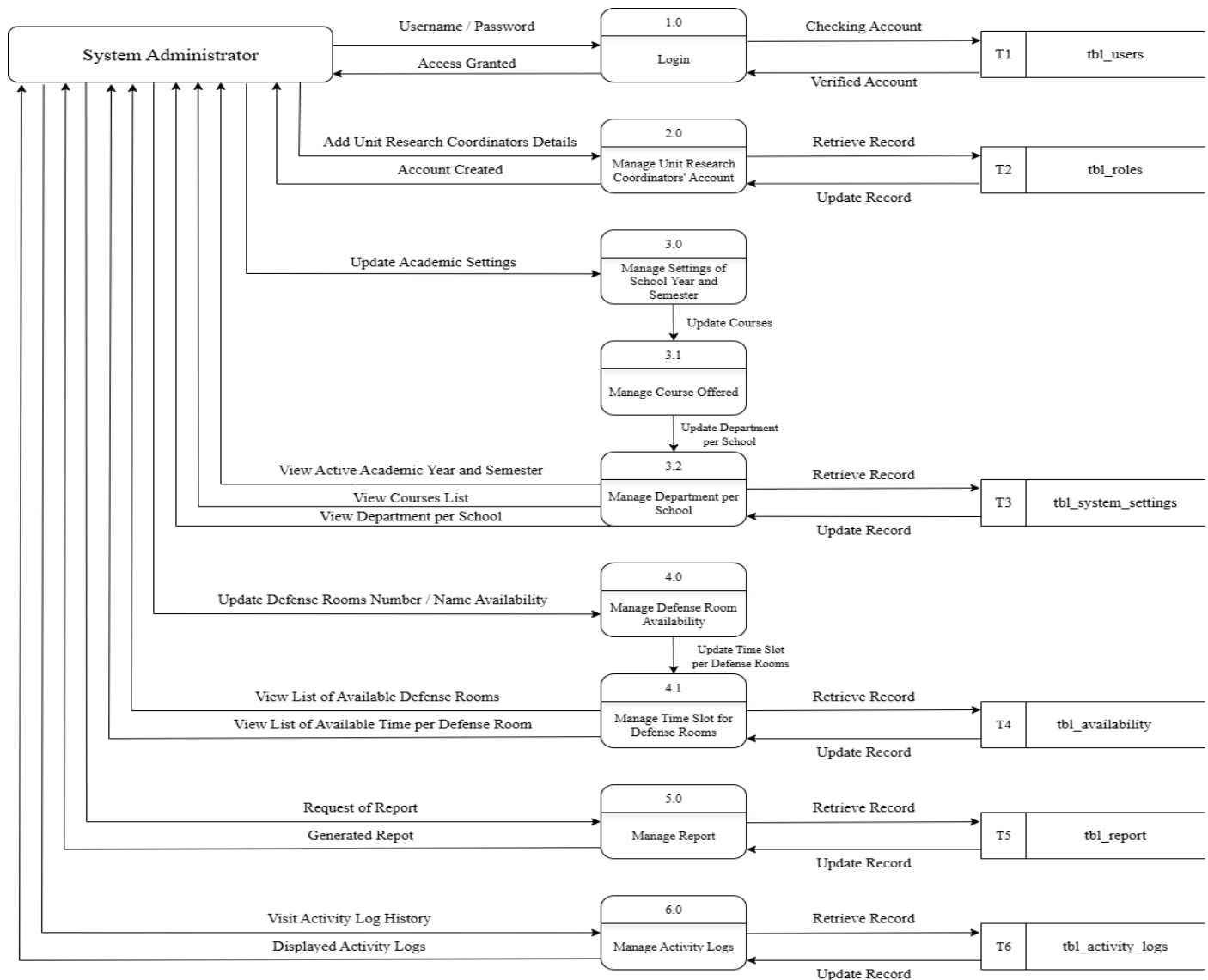


Figure 4. Data Flow Diagram of the System Administrator Entity

Figure 4 illustrates the data flow diagram of the System Administrator. The **System Administrator** within the Development of Web-Based Defense Scheduling Application for Thesis and Capstone Projects at City College of Tagaytay. The System Administrator first logs into the system, with credentials verified through the `tbl_users` table. Once authenticated, they can manage Unit Research Coordinators' accounts, referencing data from `tbl_roles`. The System Administrator

also has control over system configurations, such as managing the school year and semester settings, courses offered, and departments per school, all stored in `tbl_system_settings`. Furthermore, they oversee the defense venue by managing defense room availability and setting up time slots for these rooms, which are stored in the `tbl_availability` table. To maintain proper documentation, the administrator can generate reports from `tbl_report` and track user activities through the activity logs stored in `tbl_activity_logs`. Overall, this diagram highlights the administrator's vital role in maintaining user accounts, academic settings, defense logistics, and system monitoring, ensuring smooth and organized scheduling operations.

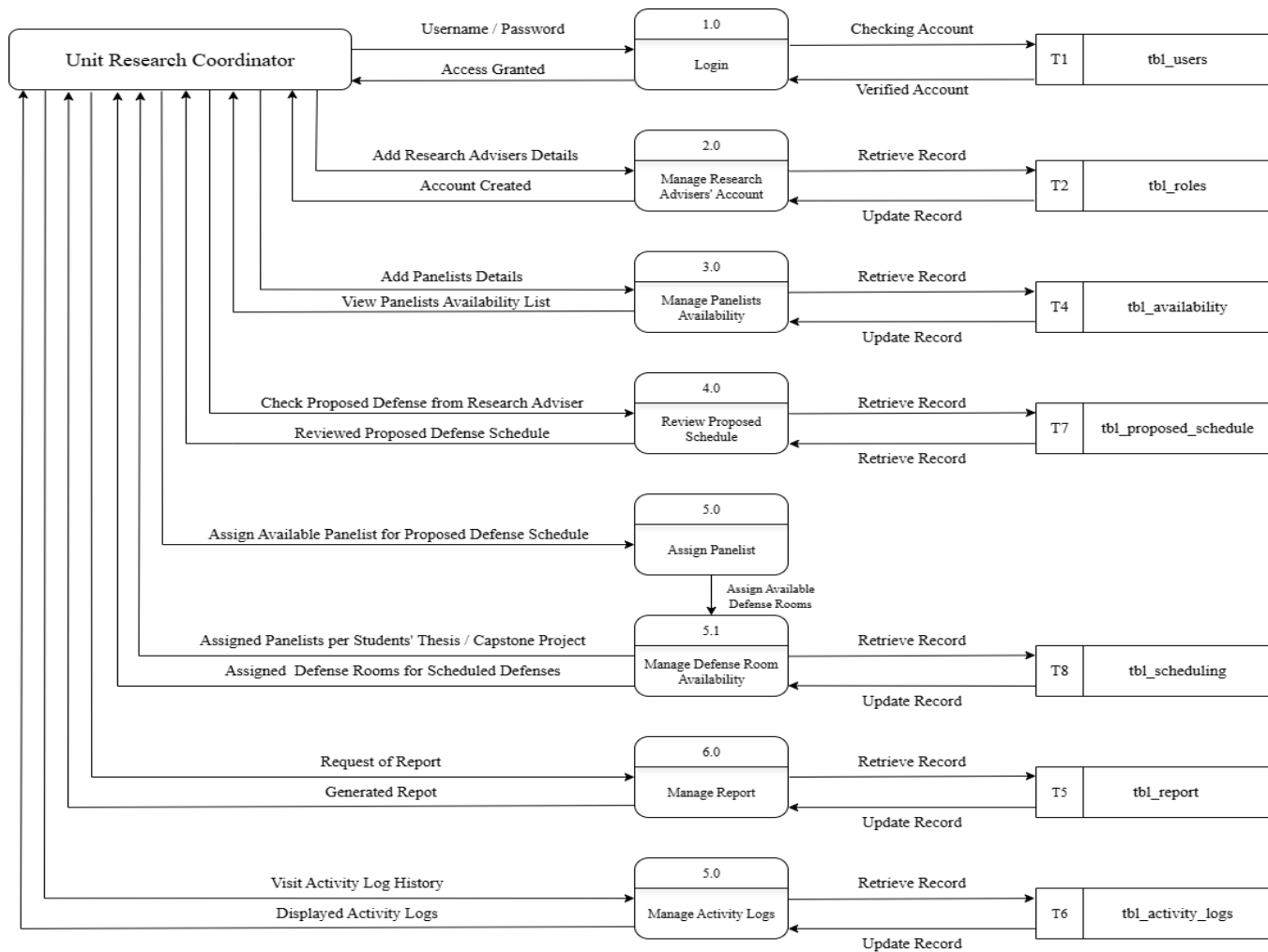


Figure 5. Data Flow Diagram of Unit Research Coordinator Entity

Figure 5 illustrates the data flow diagram of the Unit Research Coordinator. The Unit Research Coordinator within the Development of Web-Based Defense Scheduling Application for Thesis and Capstone Projects at City College of Tagaytay. The Unit Research Coordinator begins by logging into the system, where user credentials are authenticated against the `tbl_users`. Once logged in, the Unit Research Coordinator can manage research advisers' and panelist's accounts, referencing role details stored in `tbl_roles`. And also oversees panelists' availability, which is stored in the `tbl_availability` table. After the research advisers propose schedules, the Unit Research Coordinator reviews the submitted defense schedules from `tbl_proposed_schedule`. The Unit Research Coordinator are also tasked with assigning panelists and managing defense room availability, both linked to the scheduling records in `tbl_scheduling`. Additionally, the Unit Research Coordinator generates reports using data from `tbl_report` and monitors system activities through the `tbl_activity_logs`. This process ensures efficient management of accounts, schedules, availability, and documentation, reducing conflicts and maintaining transparency in the defense scheduling workflow.

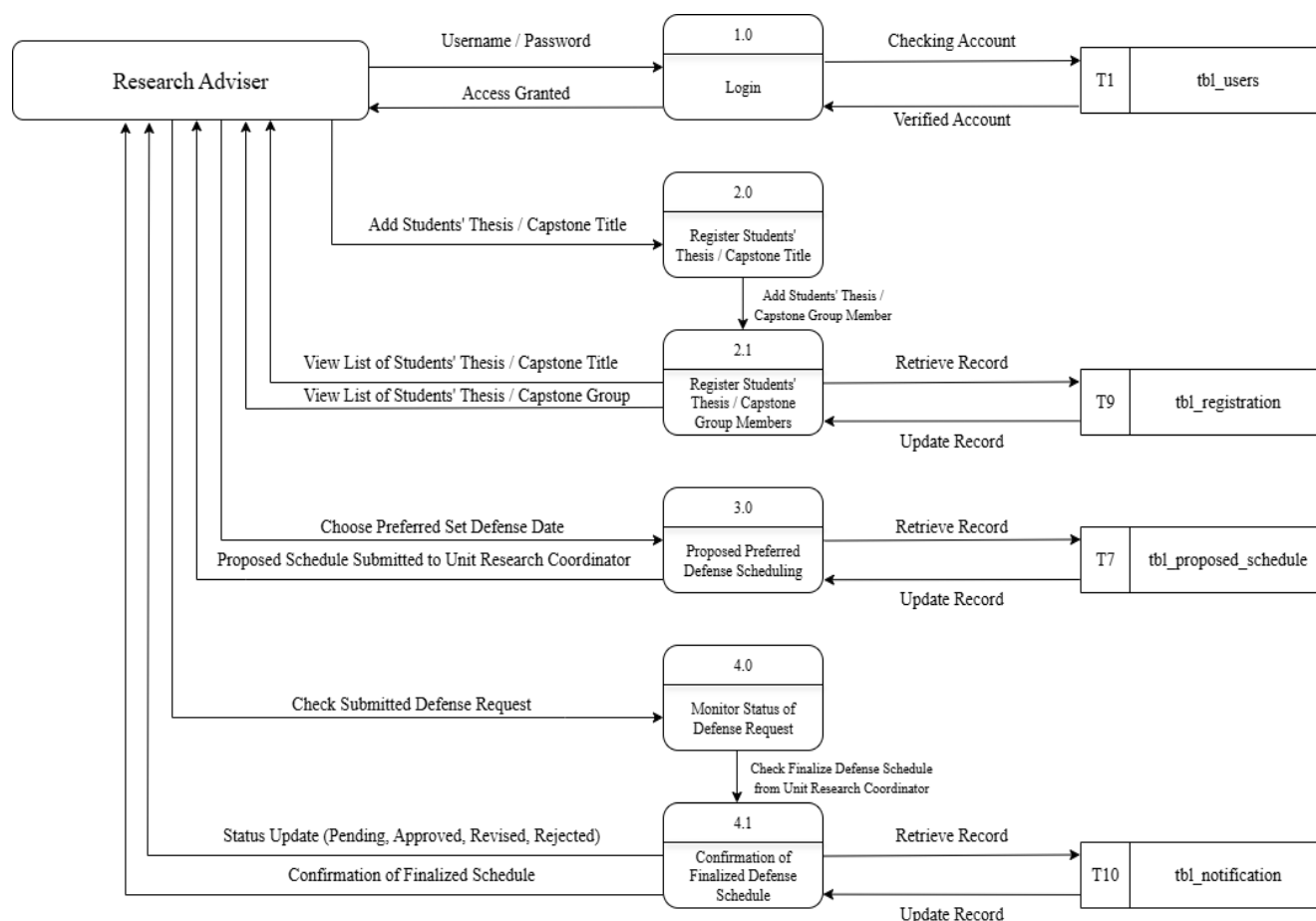


Figure 6. Data Flow Diagram of Research Adviser Entity

Figure 6 illustrates the data flow diagram of the Research Adviser. The **Research Adviser** within the Development of Web-Based Defense Scheduling Application for Thesis and Capstone Projects at City College of Tagaytay. The process begins with the adviser logging into the system, where their credentials are verified through the **tbl_users** table. Once logged in, the Research Adviser can register students' thesis or capstone titles and record group members' information, which are stored in the **tbl_registration** table. The Research Adviser also plays a role in proposing preferred defense schedules, where the details are saved in the **tbl_proposed_schedule** table. To ensure smooth coordination, advisers can monitor the status of defense requests to track approvals

and updates. Finally, once schedules are finalized, advisers confirm them, triggering notifications stored in the `tbl_notification` table. This diagram emphasizes the Research Adviser's role in facilitating student registration, managing thesis information, and coordinating defense schedules while keeping communication clear through status tracking and notifications.

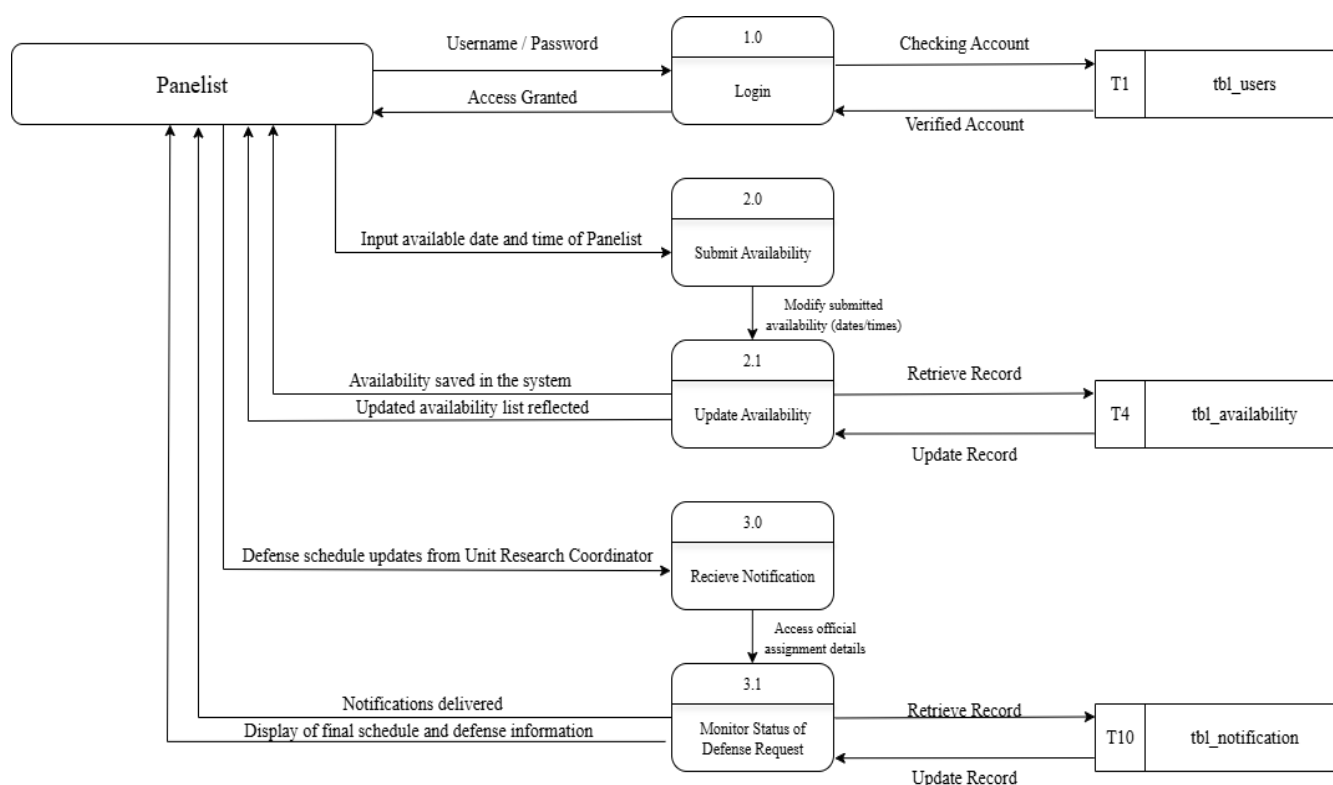


Figure 7. Data Flow Diagram of Panelist Entity

Figure 7 illustrates the data flow diagram of the Panelist within the Development of Web-Based Defense Scheduling Application for Thesis and Capstone Projects at City College of Tagaytay. The Panelist begins by logging into the system, where user credentials are verified against the `tbl_users` to ensure secure access. After logging in, the Panelist can submit their availability for defense sessions, which is stored in the `tbl_availability` table. If there are changes, the Panelist may update their availability, and the system reflects these modifications in the same

table. Once schedules are finalized, the Panelist receives notifications regarding defense assignments, which are stored and managed in the `tbl_notification` table. Additionally, the Panelist can monitor the status of defense requests through these notifications, providing them with complete details about the assigned sessions, including research titles, group members, and defense schedules. This process allows the Panelist to coordinate effectively with the scheduling system, ensuring their availability is considered while keeping them informed of their responsibilities in the defense evaluation process.

Project Methodology

The iterative methodology is a technique of project management that divides a project into phases. This is a software development approach that emphasizes cyclical development rather than trying to do everything all at once. When the work begins, researchers go through a cycle of planning, execution, and evaluation.

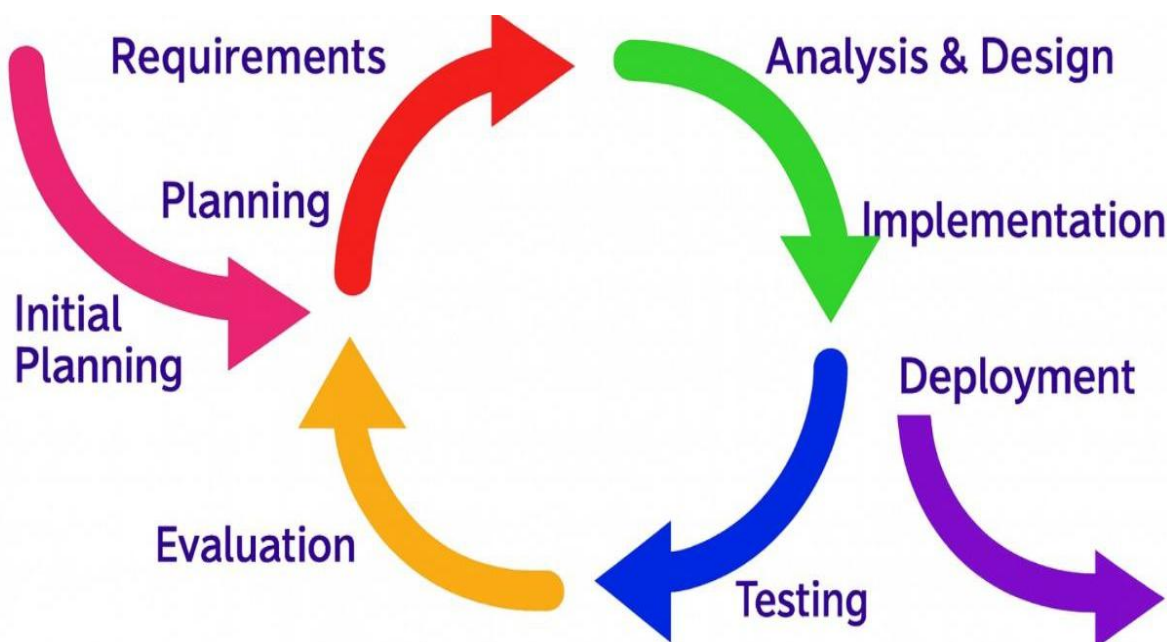


Figure 7. Iterative Methodology, Nonyelum O.F.(2020)

Figure 8 illustrates the Iterative Methodology used in Development of Web-Based Defense Scheduling Application for Thesis and Capstone Projects at City College of Tagaytay. This methodology follows a systematic cycle consisting of six (6) phases: planning and requirements gathering, analysis and design, implementation, testing, evaluation, and deployment. Each phase ensures continuous development and refinement of the system. Researchers follow this approach to effectively meet the functional and technical needs of the application while allowing room for adjustments and improvements at each stage.

Planning and Requirements Phases

First, the researchers conducted an initial planning stage in which they defined the project scope, user roles, and system requirements for the Development of Web-Based Defense Scheduling Application for Thesis and Capstone Projects at City College of Tagaytay. The team outlined the needed documentation and timeline, and gathered related information existing forms, calendars, and availability rules from the RDE Office, Unit Research Coordinators, Research Advisers, and Panelist to establish the data fields and scheduling constraints for development.

Analysis and Design Phase

The researchers analyzed the approved requirements to define the concept of the Development of Web-Based Defense Scheduling Application for Thesis and Capstone Projects at City College of Tagaytay. A context diagram was prepared to show the system boundary and its interaction with the System Administrator, Unit Research Coordinator, Research Adviser, and Panelist. Data Flow Diagrams illustrated the main processes group registration, availability/resource setup, schedule creation with conflict checking and approval, notifications,

and reports. An ERD and simple wireframes were also produced to describe the data structure and screens. These artifacts became the blueprint for implementation.

Implementation Phase

Implementation Phase began after the system design was approved and ready for development. The researchers built the system entitled “Development of Web-Based Defense Scheduling Application for Thesis and Capstone Projects at City College of Tagaytay”, following the prepared specifications and database design. Core features such as user and role management, group registration, availability setup, conflict detection, scheduling, approvals, and reporting were developed. The system was deployed in a test environment and populated with initial data, ensuring it aligned with the planned objectives before moving to the testing phase.

Testing Phase

In the testing phase for the “Development of Web-Based Defense Scheduling Application for Thesis and Capstone Project at City College of Tagaytay”, the system was presented to all intended users, including the RDE Office, Unit Research Coordinators, Research Advisers, and Panelist. The researchers tested the application to ensure user satisfaction and system reliability. This phase assessed the system’s functionality, usability, and performance, ensuring that scheduling, conflict detection, approvals, notifications, and reporting worked as intended. Feedback gathered from users during this stage was used to make necessary adjustments, ensuring the final system met the needs and expectations of its users.

Evaluation and Review Phase

This phase involves presenting the system to stakeholders for feedback and assessment. The application's functionality, usability, and performance are evaluated based on user acceptance testing and predefined quality standards. Suggestions from Research Advisers, Unit Research Coordinators, System Administrators, and Panelist are reviewed and, if necessary, integrated into the system. The evaluation also considers compliance with institutional policies and security measures to ensure readiness for actual use.

Deployment Phase

In the deployment phase of the system entitled, "Development of Web-Based Defense Scheduling Application for Thesis and Capstone Projects at City College of Tagaytay" is installed and made accessible to its intended users at the City College of Tagaytay. This may involve hosting the system on a secure web server and configuring accounts for System Administrators, Unit Research Coordinators, Research Advisers and Panelist. User training and documentation are provided to ensure smooth adoption. After deployment, the system enters the maintenance stage, where periodic updates and support are carried out to address any future issues or feature enhancements.

System Operation and Testing Procedure

This section explains how to operate the system entitled, "Development of Web-Based Defense Scheduling Application for Thesis and Capstone Projects at City College of Tagaytay" and the procedures used to test its performance. It provides clear steps for System Administrators, Unit Research Coordinators, Research Advisers, and Panelist to access and use the system, as well as the testing methods applied to ensure it is functional, reliable, and ready for use.

System Operation For System Administrator:

1. Log in with the System Administrator's credentials.
2. Configure the academic term settings, including start and end dates, defense-related deadlines, and semester details.
3. Create and manage Unit Research Coordinator accounts, assigning role-based permissions according to responsibilities.
4. Define available defense rooms, including room numbers, capacities, equipment, and time slot templates.
5. Manage and update notification settings.
6. Set up system-wide data backup schedules, password policies, and access control measures to ensure security.
7. Monitor overall system performance and review the Activity Logs for unusual or unauthorized activity.
8. Generate and export reports for administrative monitoring, evaluation, and recordkeeping.
9. Update system settings as needed to reflect changes in academic terms or institutional policies.
10. Log out once all administrative operations are complete.

For Unit Research Coordinator:

1. Log in with Research Coordinator's credentials.
2. Review academic term settings, rooms, and time slots configured by the System Administrator.
3. Create and manage Research Adviser and Panelist accounts, ensuring accurate records of their assignments and roles.

4. Review panelists' availability to build an accurate scheduling database.
5. Monitor and validate research adviser submitted student project registrations and proposed defense schedules.
6. Cross-check proposed defense schedules against panelist availability, room allocations, and academic activity calendars to prevent conflicts.
7. Assign panelists to student research groups, ensuring that advisers or designated Technical Critics/Statisticians are not assigned to their own groups.
8. Confirm available defense room and time slots, finalizing defense schedules without overlap or conflict.
9. Notify Research Advisers and Panelists of confirmed or revised schedules using the Notification Module.
10. Generate reports on schedules, assignments, and status updates for institutional recordkeeping.
11. Log out after finalizing or updating defense-related tasks.

For Research Adviser:

1. Log in with research adviser's credentials.
2. Register student project groups by entering research titles, list of members, assigned Research Adviser, and designated Technical Critic/Statistician.
3. Submit preferred defense dates and times based on group readiness, following time slots set by the Unit Research Coordinator.
4. Provide additional notes if needed, such as required equipment or special accommodations.

5. Track the status of submitted defense schedules until reviewed, revised, or finalized by the Unit Research Coordinator.
6. Confirm availability once a finalized defense schedule has been assigned.
7. Review assigned panelists and venue details, then communicate the confirmed schedule to students.
8. Respond promptly to Unit Research Coordinator inquiries or clarifications regarding the submitted details.
9. Log out after completing tasks.

For Panelist:

1. Log in with panelist credentials.
2. Submit availability details, including preferred days and times, for scheduling purposes.
3. Update availability records promptly if changes occur, ensuring accuracy in defense planning.
4. Wait for notifications regarding official defense assignments from the Unit Research Coordinator.
5. Review assigned defense schedules, including the research title, list of group members, venue, and time slot.
6. Prepare for participation in assigned defenses by reviewing project details in advance.
7. Attend scheduled defenses as assigned, fulfilling their role as an evaluator.
8. Check notifications for any updated schedules, cancellations, or reassigned sessions.
9. Log out after reviewing or completing scheduled assignments.

Testing Procedures

The system testing procedure during operation will include:

This procedure ensures that all system functions work as intended before deployment by following a step-by-step approach. Each feature of the “Development of Web-Based Defense Scheduling Application for Thesis and Capstone Projects at City College of Tagaytay” is systematically tested, with tasks executed, results observed, and errors recorded for correction. This process confirms that the system meets user requirements for accuracy, reliability, security, and overall usability, ensuring it is fully functional and ready for deployment.

Unit Testing

Unit testing focuses on verifying the functionality of individual components or modules of the “Development of Web-Based Defense Scheduling Application for Thesis and Capstone Projects at City College of Tagaytay”. Each feature, such as user login, schedule creation, or report generation, is tested separately to ensure it works correctly in isolation. This helps identify and fix errors early in the development process before they affect other parts of the system.

Integration Testing

Integration testing ensures that different modules of the system work together seamlessly. In this phase, connected features such as the interaction between the scheduling module and the user management module are tested to confirm proper data flow and communication. This step helps detect issues that may arise when modules exchange information or operate in combination.

System Testing

System testing phase, involves evaluating the entire “Development of Web-Based Defense Scheduling Application for Thesis and Capstone Projects at City College of Tagaytay” as a whole. It checks whether all integrated components meet the defined requirements and function correctly in the intended environment. This includes testing usability, performance, security, and compatibility to ensure the system delivers a smooth and reliable experience for all types of users.

Acceptance Testing

Acceptance testing is the final stage, where the system is tested by actual end users such as system administrators, unit research coordinators, research advisers, and panelist to verify that it meets their needs and expectations. This step ensures the application aligns with the requirements set at the start of the project. Successful acceptance testing confirms that the system is ready for deployment and real-world use at the City College of Tagaytay.

Evaluation Procedure

The evaluation procedure for the Development of Web-Based Defense Scheduling Application for Thesis and Capstone Projects at City College of Tagaytay focuses on assessing the system’s functionality, usability, reliability, and overall effectiveness. During the preliminary evaluation, the researchers tested the prototype to ensure it met the specified system requirements, checking for correct processing of data, proper scheduling functions, and accurate output generation. For the final evaluation, feedback was collected from key users, including system administrators, unit research coordinators, research advisers, and student, through structured evaluation sheets, surveys, and direct observation of system use. The comments, ratings, and

suggestions provided by users were analyzed to identify strengths, weaknesses, and areas for improvement. The evaluation criteria were based on the ISO 25010 Standard for software quality, with a rating scale from 1 to 5, where 1 indicates poor performance and 5 indicates excellent performance, to systematically measure the system's effectiveness and guide further enhancements.

Table 1. System Evaluation Sheet Numerical and Descriptive Scale

NUMERICAL RATING	INTERPRETATION	DEFINITION
4.21 - 5.0	Excellent	The system fully meets and far exceeds most expectations.
3.41 - 4.20	Very Good	The system fully meets all and exceeds several expectations.
2.61 - 3.0	Good	The system fully meets all expectations.
1.81 - 2.60	Fair	The system does not fully meet all expectations.
1.00 - 1.80	Poor	The system fails to meet expectations to a significant degree in several areas.

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